RESEARCH MANUSCRIPT / REPRODUCIBLE COMPUTATIONAL BASELINE

PLANETARY COUPLED PULSE THEORY

PCPT v3.0

Mathematical & Experimental Validation

Geological Rectification Mechanism (GRM)

Conservation-controlled benchmark | Quantitative predictions | Field-test protocols



SCIENTIFIC STATUS — MODEL VERIFIED, NATURAL MECHANISM NOT VALIDATED.
The present edition contains an independently executable *synthetic* analytical and finite-volume numerical benchmark. No new laboratory, borehole or field measurements were supplied or collected. Published observations are cited only as external prior art. The predicted rectified flow is conditional on a hypothetical constitutive law and imposed boundary forcing; it is not evidence of a planetary thermal circuit.

Theory author and system architect: Vadym Tsinderhoz

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Publication record and accountability

Record	Value
Project	Planetary Coupled Pulse Theory (PCPT)
Theory author	Vadym Tsinderhoz
Edition	v3.0 - Mathematical & Experimental Validation
Prior baselines	PCPT v1.0 (131 pages, 28 July 2026); Corrected Scientific Edition v2.1 (144 pages, 28 July 2026).
Current publication date	2 November 2026
Scientific contribution	A sharply bounded nonlinear-Darcy GRM benchmark, exact cycle-mean solution, finite-volume conservation test, thermal screening, explicit validation and stopping gates.
Not provided	New raw field data, core-laboratory measurements, independent experimental replication, demonstrated natural values of the pressure-sensitivity parameter.
Reproduction	Supplementary Python code, model inputs, CSV outputs and figures in the companion archive.
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Evidence classification

Class	Meaning within this manuscript
E1 — Established	Independently documented physical effect or governing equation.
E2 — Published observation	Specific third-party laboratory/field/satellite result; not new PCPT data.
M — Verified model result	Output generated by the attached synthetic model under declared assumptions; not a natural observation.
H — Hypothesis	Proposed mechanism requiring controlled measurement and independent replication.
X — Excluded	An interpretation explicitly not supported by the model or sources.

Boundary of claims. PCPT v3.0 does not establish H-04 or H-05 in real geology, does not model petroleum circulation, and does not infer climate-scale forcing from local fluid transport. Simulation is verification of equations and implementation; validation requires observations not used to construct the model.

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1. Executive summary

Research question. Can a zero-mean periodic pressure applied to a porous or fractured geological system with an admissible nonlinear hydraulic constitutive relation generate a reproducible, non-zero cycle-mean Darcy flux while satisfying local and integral mass conservation? If so, under what conditions would this produce an independently measurable incremental thermal effect?

Scope. This edition isolates the GRM branch of PCPT v2.1, especially H-04 (rectified fluid flow) and H-05 (added heat transport), from the much broader Sun–Moon–Earth architecture. The experiment uses a *prescribed periodic boundary pressure*, not a computed Earth-tide-to-pore-pressure transfer. The forcing magnitude is therefore illustrative and must not be attributed to a given natural site without measurements.

Model. A saturated, isothermal hydraulic interval of length L is governed by single-phase Darcy flow with $k(p) = k_0 \exp(\beta p)$ and constant storage C. Two controlled boundary pressures drive a periodic flow. An exact quasi-steady cycle-mean solution provides an analytic benchmark; a nonlinear implicit finite-volume solver independently tests transient response and mass closure. A subsequent steady advection–conduction calculation estimates thermal consequences of the *mean* flux, explicitly ignoring time-dependent thermal transients.

Actual synthetic result. For the declared demonstration case ($L = 10$ m, $k_0 = 10^{-13}$ m^2, $\mu = 10^{-3}$ Pa s, $\beta = 2 \times 10^{-5}$ Pa^{-1}, $A = 10$ kPa, $P = 12.42$ h), the analytical time-mean is $5.01251390 \times 10^{-9}$ m/s, or 0.4331 L day^{-1} per m^2 of flow area. A 32-cell, ten-cycle finite-volume calculation returns $5.01251413 \times 10^{-9}$ m/s at the outlet, with relative disagreement 4.65×10^{-8}. The same solver with $\beta = 0$ gives a numerically zero cycle-mean. These values are *synthetic only*.

Thermal screen. A 0.30 K temperature difference over 10 m gives Pe $= 0.1053$. With prescribed end temperatures, the boundary total heat-flux change relative to conduction-only is about $+0.00310$ W/m^2 in the specified sign convention. It is not global heat flux, nor a measured extraction effect.

Core conclusion. The equations and code demonstrate that a specifically defined nonlinear Darcy system *can* rectify an imposed zero-mean boundary pressure. They do not establish that real formation constitutive laws have the required parameters, that tidal forcing supplies the assumed pressure, or that a measurable natural signal exceeds competing sources. The decisive next step is a pre-registered and independently instrumented laboratory test, not global extrapolation.

2. Lineage, claim boundary and scientific novelty

2.1 Traceability to prior PCPT editions

PCPT v1.0 proposed a broad integrated research architecture and a 24-hypothesis catalogue. The corrected v2.1 narrowed its core experimental claims to ten named hypotheses, including H-03 (pressure response to loading), H-04 (non-zero cycle mean), H-05 (incremental heat transport) and H-08 (scaling). This edition makes the H-04 numerical benchmark executable while retaining H-05 as a thermal *screen* rather than a field result.[1]

Claim	Current object	v3.0 disposition
H-03	External loading to pore pressure	Prescribed boundary condition only; no geomechanical forcing validation.
H-04	Rectification	Analytic and nonlinear finite-volume *model demonstration*; natural existence unresolved.
H-05	Incremental heat advection	Mean-flux conduction–advection screening calculation; no thermal data validation.
H-06	Extraction/injection change	Study design and metadata requirements; not simulated.
H-08	Regional/global significance	Stop-rule framework only; no global extrapolation performed.

[1] Source documents: PCPT v1.0, pp. 3–6 and 114–116; PCPT v2.1, pp. 3, 43–45, 137–140 and 144.

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2.2 What may be new, and what is prior art

Earth-tide-driven groundwater response is established, with known wellbore, aquitard and parameter-inference limitations [R2,R5,R6]. A separate controlled Hele–Shaw analogue already demonstrated time-averaged tidal groundwater circulation in a coastal configuration [R3]. Therefore "periodic forcing can generate groundwater circulation" is *not* a novel first observation. The potentially researchable PCPT contribution is a specific, independently predictive relation between *measured* stress-dependent permeability, oscillatory forcing, controlled net transport and additional heat transport across selected geological configurations. Establishing priority or novelty beyond these papers requires an exhaustive scholarly and patent prior-art assessment not claimed here.

3. Prior empirical evidence and the missing dataset

Record	Established or reported observation	Relation to PCPT / principal limitation
R2; USGS (1989)	Water-level response in five wells to Earth tides and atmospheric loading, interpreted with poroelastic formation properties.	Supports a loading-to-well-response channel. Does *not* demonstrate time-mean throughflow, because well water levels alone cannot quantify integrated transport.
R3; *Geology* (2004)	Scaled Hele–Shaw laboratory analogue shows asymmetric tidal exchange and a time-averaged coastal groundwater circulation.	Relevant prior example of rectification. Coastal unconfined geometry differs from the proposed confined 1D constitutive test.
R4; *Science* (2003)	Satellite magnetic measurements identify M2 ocean-tide induction.	Supports one upstream PCPT physical channel; unrelated to direct proof of GRM.
R5; *JGR* (2022)	Numerical examination of Earth-tide-driven well response identifies restricted validity and inversion sensitivity.	Strong warning against inferring permeability or net flow from two harmonic observables without appropriate geometry and storage.
R6; *HESS* (2024)	Leaky-aquifer groundwater tide model includes aquitard storage, wellbore storage and skin; compared with measurements.	Identifies indispensable nuisance mechanisms for future field tests.

PCPT v3.0 raw evidence register: natural measurements = **0 new records**; newly analysed laboratory cores = **0**; newly acquired well-flow series = **0**. Published observations above are literature evidence, not datasets reanalysed in this edition. The CSV generated by the companion code is simulated and explicitly marked as such.

4. Definitions, signs and control volume

Let $x \in [0, L]$ denote distance measured from a driven upstream pressure reservoir toward a fixed-pressure outlet. Positive Darcy flux q denotes material moving toward increasing x. Let $p(x, t)$ be gauge fluid pressure (Pa) relative to the fixed right boundary; C denotes effective specific volumetric storage per bulk volume per unit pressure (Pa^{-1}). The single-phase, saturated medium has fixed viscosity μ and pressure-dependent absolute permeability $k(p)$.

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Symbol	Unit	Operational meaning
x, L	m	Axial coordinate and specimen/reservoir interval length.
p, p_L, p_R, A	Pa	Gauge pore pressure, left/right boundary values, forcing amplitude.
q	m s^{-1}	*Darcy* or superficial volume flux per gross cross-section, not pore velocity.
$k_0, k(p)$	m^2	Reference and pressure-sensitive intrinsic permeability.
β	Pa^{-1}	Logarithmic sensitivity $\partial \log k/\partial p$ of the hypothetical constitutive relation.
μ	Pa s	Fluid dynamic viscosity.
C	Pa^{-1}	Effective storage coefficient in this reduced 1D model.
T_f, P	K, s	Temperature field and period of forcing.
$\lambda, \rho c_p$	W m^{-1} K^{-1}; J m^{-3} K^{-1}	Effective thermal conductivity and volumetric fluid heat capacity.

Pore speed is $v_p = q/\phi$ only for an appropriate connected pore fraction ϕ; the benchmark uses q and does not infer v_p. Gravity and buoyancy are excluded by choosing a horizontal interval and equal elevation of boundaries; poroelastic stress-to-pressure forcing, multiphase flow, viscoelastic hysteresis, fracture displacement, and chemistry are likewise outside the reduced solver.

5. Minimal mathematical formulation

5.1 Pressure-dependent permeability and Darcy law

The deliberately falsifiable constitutive hypothesis is

$$k(p) = k_0 \exp\left[\beta(p - p_{\text{ref}})\right], \qquad p_{\text{ref}} = 0. \tag{1}$$

This local exponential law is a *candidate approximation over a bounded pressure range*, not a universally established law for rock. Its derivative at reference conditions is $dk/dp = \beta k_0$. The Darcy flux and fluid-volume continuity in the reduced isothermal model are

$$q(x,t) = -\frac{k(p)}{\mu}\frac{\partial p}{\partial x}, \tag{2}$$

$$C\frac{\partial p}{\partial t} + \frac{\partial q}{\partial x} = 0. \tag{3}$$

Boundary and initial conditions are

$$p(0,t) = A\sin(2\pi t/P), \qquad p(L,t) = 0, \qquad p(x,0) = 0. \tag{4}$$

The mean imposed difference $\langle p_L - p_R \rangle = 0$. The driving boundary is an ideal pressure reservoir able to supply/extract fluid and mechanical work. No spontaneous energy generation is implied.

5.2 Exact cycle-mean expression

Introduce the Kirchhoff potential

$$F(p) = \int_0^p \frac{k(s)}{\mu}\, ds = \begin{cases} \frac{k_0}{\mu\beta}\left(e^{\beta p} - 1\right), & \beta \neq 0, \\ \frac{k_0}{\mu} p, & \beta = 0. \end{cases} \tag{5}$$

Then $q = -\partial F/\partial x$. Over a periodic stationary cycle, integrating conservation over time yields $\partial_x \bar{q} = 0$, and hence *without requiring an instantaneous quasi-steady interior*,

$$\bar{q} = \frac{\langle F(p_L)\rangle - \langle F(p_R)\rangle}{L} = \frac{k_0}{\mu\beta L}\left[I_0(\beta A) - 1\right], \tag{6}$$

where I_0 is the modified Bessel function of order zero. This average formula follows from the stated 1D constitutive relation and periodic steady state; it is *not* proof that the boundary process exists at natural sites. The instantaneous quasi-steady flux $q(t) = [F(p_L(t)) - F(p_R(t))]/L$ is a separate stronger approximation and is used only for an illustrative waveform and quadrature check.

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For $|\beta A| \ll 1$,

$$\bar{q} \approx \frac{k_0 \beta A^2}{4\mu L} + O(\beta^3 A^4). \tag{7}$$

This predicts a sign reversal with β, an approximately quadratic amplitude law, and a zero mean for the linear null $\beta = 0$. Any measured deviation must be evaluated against background pressure gradients and boundary conditions rather than immediately attributed to a new mechanism.

5.3 Storage and physically closed boundaries

The experimental system must possess hydraulically open boundary reservoirs to measure throughflow. For an impermeable (no-flow) closed system, the cycle-mean boundary flux *must be zero* by mass conservation irrespective of $k(p)$. Therefore the present benchmark cannot imply a one-way flow around a closed, sealed planetary system. It represents a forced open interval with continuous exchange. In laboratory tests, *both reservoirs' volumes and pumps must be metered*; pressure oscillation alone is insufficient.

6. Finite-volume solver and verification methodology

Let N equally sized control volumes have center pressures p_i^n and width $\Delta x = L/N$. The face flux uses the exact Kirchhoff potential difference

$$q_{i+1/2}^{n+1} = -\frac{F(p_{i+1}^{n+1}) - F(p_i^{n+1})}{\Delta x}, \qquad q_{1/2}^{n+1} = -\frac{F(p_1^{n+1}) - F(p_L^{n+1})}{\Delta x/2}. \tag{8}$$

The outlet face is defined symmetrically, with boundary-to-center distance $\Delta x/2$. Backward Euler solves

$$C\Delta x \frac{p_i^{n+1} - p_i^n}{\Delta t} + q_{i+1/2}^{n+1} - q_{i-1/2}^{n+1} = 0. \tag{9}$$

A tridiagonal Newton iteration applies the exact derivative $F'(p) = k(p)/\mu$. Every update checks the discrete volume budget

$$\Delta \mathcal{S} = \sum_i C(p_i^{n+1} - p_i^n)\Delta x = \left(q_{\text{in}}^{n+1} - q_{\text{out}}^{n+1}\right)\Delta t. \tag{10}$$

The reference implementation runs $N = 32$ cells, ten periods and $\Delta t = 119.8713$ s (373 steps per period). The initial state is uniform $p = 0$. Only the last two periods are exported to CSV. Convergence is checked against an exact analytic mean and a linear null. There are no random seeds or hidden calibration parameters.

Verification metric	Executed result
Exact analytic cycle-mean $\bar{q}$	$5.0125138976 \times 10^{-9}$ m/s
Finite volume, outlet time-mean (last cycle)	$5.0125141308 \times 10^{-9}$ m/s
Relative difference from exact value	4.65×10^{-8}
Difference: inlet and outlet cycle-mean	-4.41×10^{-15} m/s
Maximum absolute mass-closure residual per step	5.07×10^{-11} m (equivalent liquid height)
Finite-volume linear null, $\beta = 0$	-9.08×10^{-24} m/s

Table 2: Outputs from executing the supplied Python benchmark on 2 November 2026. They validate internal numerical consistency only.

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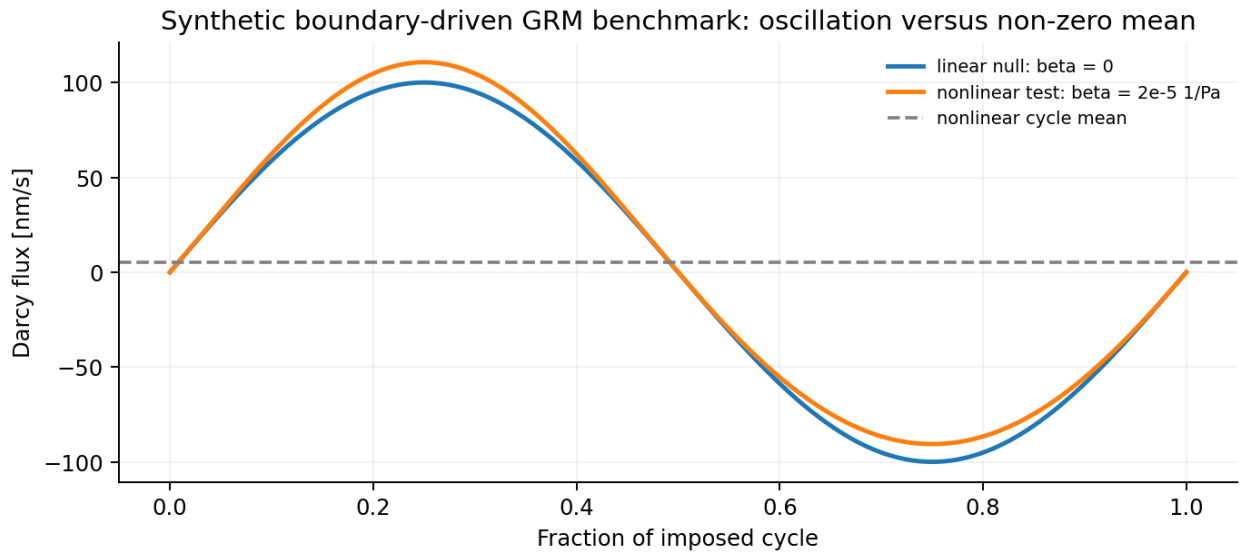


Figure 1: Model-derived instantaneous quasi-steady flux. Only the nonlinear constitutive law produces a non-zero cycle mean; no measured data appear in this figure.

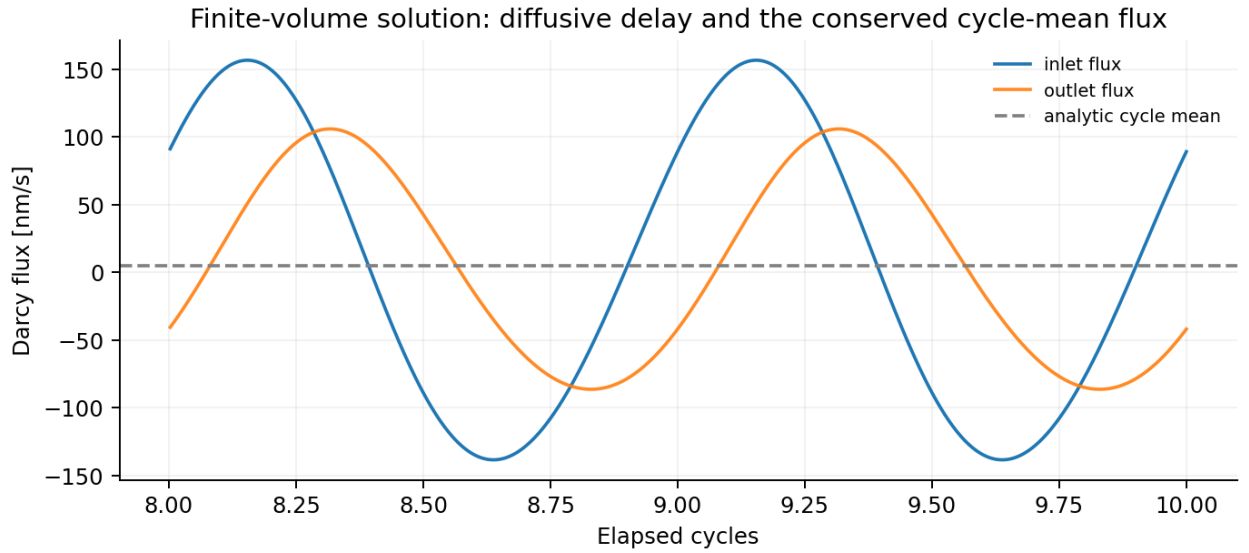


Figure 2: Implicit finite-volume transient solution for the last two periods. Phase differences reflect storage and hydraulic diffusion. Outlet and inlet fluxes have equal cycle means after equilibration.

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7. Declared synthetic parameter case

Parameter	Demonstration value	Interpretation and caution
L	10 m	Laboratory/idealized hydraulic interval; not a planet-wide path.
k_0	10^{-13} m^2	Assigned reference permeability; not measured from any core.
μ	10^{-3} Pa s	Water-like viscosity for this synthetic case.
C	2×10^{-8} Pa^{-1}	Assigned storage; used in time-dependent solution.
β	2×10^{-5} Pa^{-1}	Assumed stress sensitivity: $k(\pm 10\,\text{kPa})/k_0 = e^{\pm 0.2}$. *No geological calibration.*
A	10 kPa	Ideal boundary oscillation amplitude, not an asserted Earth-tide pore pressure.
P	12.42 h	Illustrative semidiurnal tidal-like period, not verified local harmonic forcing.
ρc_p	4.2×10^6 J m^{-3} K^{-1}	Assigned water volumetric heat capacity.
λ	2.0 W m^{-1} K^{-1}	Assigned effective conductivity.
ΔT	0.30 K	Prescribed 10-m temperature contrast (0.03 K/m).

Table 3: All values used for generated figures and quantitative outputs. These parameters are choices for demonstration, not estimates of any specific site.

A pressure perturbation of 10 kPa is an idealized, relatively large driving boundary perturbation. H-03 must first demonstrate that any real tidal or atmospheric loading can generate a comparable *formation pressure* at the candidate site. This is the major unresolved link between the broader PCPT forcing chain and the present hydraulic benchmark.

8. Quantitative and falsifiable predictions

The model yields *conditional* predictions that an experiment can test after independently calibrating k_0, β, C and the boundary pressures:

1. **Zero-response null.** If $\beta = 0$ and the setup has no background gradient or asymmetric boundary exchange, the cycle mean must vanish up to measurement uncertainty.
2. **Sign law.** With the pressure reference and flow direction stated above, $\text{sgn}(\bar{q}) = \text{sgn}(\beta)$ for nonzero β and A.
3. **Amplitude law.** For weak nonlinearity $|\beta A| \ll 1$, $\bar{q}$ scales as A^2 rather than A. Doubling amplitude should approximately quadruple the rectified component if the same constitutive regime holds.
4. **Mass constraint.** At periodic stationarity, inlet and outlet *mean* Darcy fluxes must agree after known leakage and storage changes are included.
5. **Frequency check.** In the narrow model with constant C and fixed $k(p)$, the exact *mean* (not the instantaneous phase response) is independent of period P. Reproducible frequency dependence of the mean signals additional physics such as evolving aperture, time-dependent chemistry or changing boundaries.
6. **Baseline-pressure sensitivity.** Adding a background gradient changes the null and must be included explicitly; no measured nonzero mean is diagnostic by itself.

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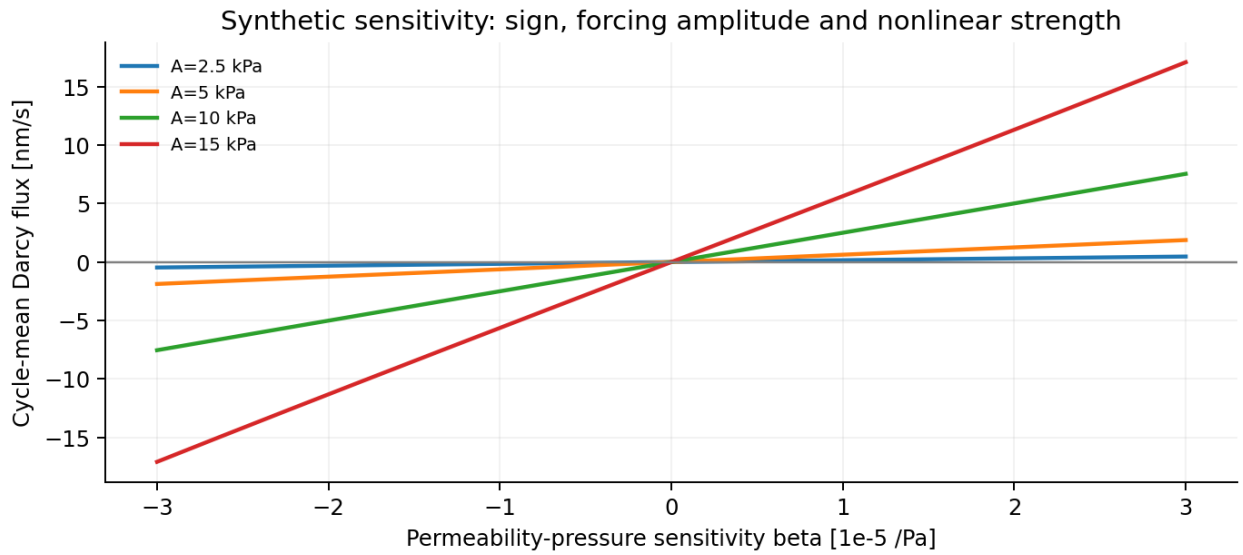


Figure 3: Sensitivity of the exact synthetic mean flow to β and applied pressure amplitude. The curve is a model prediction, not an empirical parameter estimate.

Derived quantity, demonstration case	Value
βA	0.200 (dimensionless)
$k(+A)/k_0$ and $k(-A)/k_0$	1.2214 and 0.8187
Time-mean flux $\overline{q}$	5.0125 nm/s
Throughput across 1 m^2 gross area	0.4331 L/day
One-year equivalent Darcy displacement	0.1582 m/year
Mean boundary-pressure work, quasi-steady wave	5.025×10^{-4} W/m^2

Table 4: The annual displacement is $q\times$ time, not a pore-water particle path. Pressure work is calculated from $\langle p_L q_{QS} \rangle$ and is not a field energy budget.

9. Thermal screening: H-05 is not yet verified

For a one-dimensional *steady mean-flow approximation*, the fluid heat equation with fixed end temperatures and no volumetric sources is

$$\rho c_p \overline{q} \frac{dT}{dx} = \lambda \frac{d^2T}{dx^2}, \qquad T(0) = T_0, \quad T(L) = T_0 + \Delta T. \tag{11}$$

The Peclet number and exact profile are

$$\mathrm{Pe} = \frac{\rho c_p \overline{q} L}{\lambda}, \tag{12}$$

$$\frac{T(x) - T_0}{\Delta T} = \frac{e^{\mathrm{Pe}\, x/L} - 1}{e^{\mathrm{Pe}} - 1}, \qquad \mathrm{Pe} \neq 0. \tag{13}$$

The *total*, reference-enthalpy heat flux is $J = \rho c_p \overline{q}(T - T_0) - \lambda\, dT/dx$. Evaluated at $x = 0$ for the example parameters,

$$\mathrm{Pe} = 0.1052628, \tag{14}$$

$$J_0 \text{ (conduction only)} = -0.060000 \text{ W m}^{-2}, \tag{15}$$

$$J \text{ (mean-flow model)} = -0.056898 \text{ W m}^{-2}, \tag{16}$$

$$\Delta J = +0.0031025 \text{ W m}^{-2}. \tag{17}$$

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The naive transport scale $\rho c_p \bar{q} \Delta T = 0.006316$ W/m^2 is *not* itself the change in total heat flux because the conductive gradient readjusts. The sign and magnitude depend on which boundary is hotter, the direction of flow and the enthalpy reference. A rigorous future test must solve the time-dependent advective–dispersive energy equation for the measured velocity history, account for solid/fluid heat capacities, and measure inlet/outlet heat.

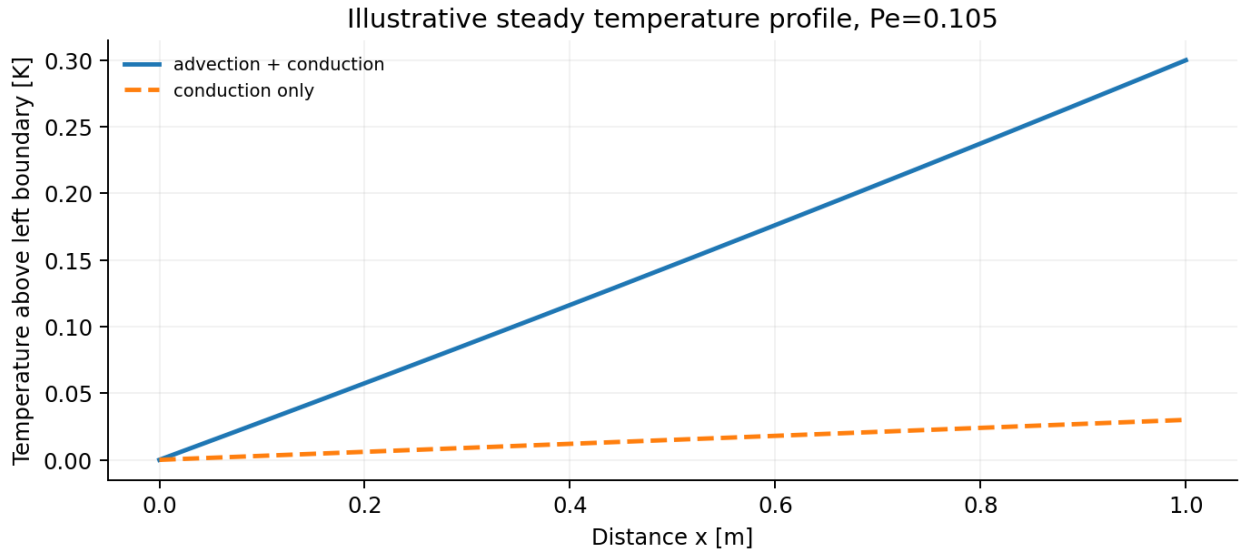
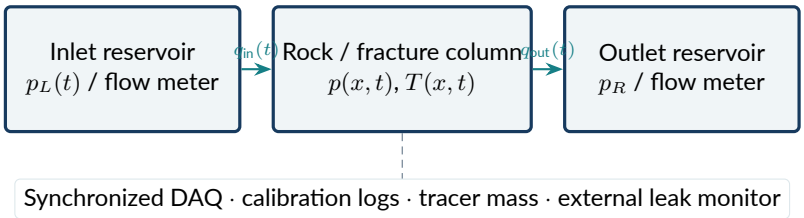


Figure 4: Mean-flow steady thermal screen for the prescribed 0.30-K end-to-end contrast. The difference is not a measured geological temperature anomaly.

10. Proposed laboratory experimental programme

10.1 Apparatus and control volumes

Construct a saturated flow column with measured L and cross-section A_c, two independent pressure-controllable reservoirs, positive-displacement flow sensing on *both* ends, differential pressure transducers along the column, temperature probes, inlet/outlet tracer sampling and an external leak-monitoring circuit. Calibrate dead volume, tubing compliance, reservoir pressure response, bypass leakage, pump stroke bias and fluid thermal expansion before applying cyclic forcing. A rigid reference column provides the linear null; a second assembly contains independently characterized pressure-sensitive fractures or compliant porous material.



10.2 Pre-registered trial sequence

1. **T0: Stationary calibration.** Determine baseline hydraulic conductivity, pressure-dependent permeability $k(p)$ over the entire proposed range, storage, leak rate and flow-sensor offsets using independent static and step-pressure tests; do not fit them with cyclic-run data.
2. **T1: Linear/null control.** Apply periodic zero-mean boundary forcing to a rigid/control column across multiple amplitudes and periods; verify $\bar{q}$ is consistent with zero and with instrument limits.
3. **T2: Nonlinear column.** Randomize conditions across amplitude, frequency, baseline pressure and orientation. Capture at least 30 complete cycles per treatment *as an initial planning target*; revise the sample size

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from pilot-estimated autocorrelation and variance before confirmatory preregistration.

4. **T3: Direct transport confirmation.** Reconcile time-integrated inlet/outlet volume and a tracer mass balance. Separate any apparent mean from cyclic tubing storage or temperature expansion.
5. **T4: Thermal contrast.** Repeat with independently maintained boundary temperatures, collect distributed thermometry and calorimetry; compare predicted total heat flux with a conduction-only experiment at matching pressures.
6. **T5: Independent duplication.** Another team repeats the locked procedure with a second test assembly; methods, negative outcomes and parameter uncertainty are published.

10.3 Operational acceptance criteria — proposed, not achieved

Primary estimate: $\overline{q}_{\text{observed}}$ from the last pre-specified stable cycles using integrated volume per period and gross flow area. Primary null: a model with $\beta = 0$ and measured, explicitly corrected background drift. Preregistration must fix the test statistics, sign convention, intervals and exclusion criteria *before* the confirmatory run. A candidate H-04 signal is accepted only if it reproduces the predicted sign and amplitude response, is non-zero under a 95% uncertainty interval accounting for time dependence, passes tracer/leak control and is replicated on a second assembly. These are evidence *gates*, not arbitrary p-value guarantees.

11. Field validation: site selection and source separation

Field evaluation begins only after measured permeability sensitivity and lab rectification can bound the expected signal. Preferred design includes (i) a control well unaffected by pumping, (ii) a monitored fractured or coastal formation with pressure and deformation channels, and (iii) a well with known extraction/injection events. The field group must record barometric pressure, ocean/earth-tide constituents from official ephemerides, precipitation, aquifer levels, well completion, pump operations, fracture orientation, conductivity/temperature logs, and GNSS or strain, where feasible. Data sources for potential future ingress include IERS Conventions [R7], NOAA CO-OPS [R8], USGS groundwater and well metadata [R9]; *no records from those sources are ingested as validation observations in this edition*.

11.1 Observation and nuisance model

For well j, define a recorded pressure or head observation $y_j(t)$ and a calibrated hydraulic prediction $m_j(t;\theta)$. Model

$$y_j(t) = m_j(t;\theta) + b_j\big(P_{\text{atm}}(t), u_{\text{pump}}(t), h_{\text{water}}(t), T_{\text{well}}(t)\big) + d_j(t) + \epsilon_j(t), \tag{18}$$

with explicit nuisance function b_j, sensor drift d_j, and residual ϵ_j (including autocorrelation). Wellbore storage/skin and leaky aquitard response should be estimated using dedicated tests, not ignored [R5,R6]. A spectral peak coincident with M2, S2, O1 or K1 is *not* evidence for a causal throughflow without a measured fluid mass transfer and valid confounder controls.

12. Statistical design and identifiability

- **Primary estimand**: net volume per cycle normalized by gross area and period, with signed direction defined before inspection.
- **Separate training and confirmation**: calibrate k_0, β, C on static/step tests and dedicated calibration cycles; lock parameters for confirmatory periodic experiments.
- **Autocorrelation**: analyse per-cycle integrated volumes and form uncertainty intervals via an appropriate block bootstrap or hierarchically specified correlated-error model; report effective sample count, not raw sensor sample count.
- **Minimum detectable effect**: set after a blind pilot using observed drift, flux noise and leakage; avoid asserting that the synthetic 0.433 L/day/m^2 is detectable with unspecified instruments.
- **Controls**: baseline-pressure-only, zero-amplitude, reversed orientation, rigid reference, pump-off and dummy sensor channels; randomize test order where practical.

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- **Model comparison**: compare fixed-parameter nonlinear model versus independently calibrated linear, leak-only, thermal-expansion and equipment-compliance alternatives. Use held-out predictive scores, not only correlation.
- **Multiple testing**: preregister a single primary H-04 outcome; H-05 is a separate later experiment. Exploratory sites, frequencies and lags do not count as confirmatory outcomes.

13. Verification and validation gates

Gate	Objective	Minimum evidence to advance	Status
G0	Dimensions, scaling	SI-consistent equations; physically admissible boundary/work budget; sensitivity range declared.	Partial
G1	Numerical verification	Exact benchmark, zero-mean linear null, mass balance, export of executable code and CSV.	Pass*
G2	Literature/provenance	Cite direct prior art, separate primary observations from model-generated outputs.	Partial
G3	Laboratory flow H-04	Independent permeability characterization; volume and tracer balance; nonzero replicated flow after controls.	Not done
G4	Thermal H-05	Time-resolved temperatures and calorimetric energy closure; improvement over conduction-only model.	Not done
G5	Field H-03/H-04	Calibrated transfer from astronomical/mechanical loading to true hydraulic flow, independent sites.	Not done
G6	Regional significance	Representative spatial power integration with uncertainty and plausible external energy supply.	Not done

*G1 is a statement of consistency for the supplied reduced model, not external code certification, field validation or independent formal proof. A true independent audit remains outstanding.

14. Requirements traceability and uncertainty budget

ID	Requirement	Verification / evidence artifact
REQ-V3-01	Declare each SI unit, spatial domain and sign convention.	Section 4; code parameter metadata.
REQ-V3-02	Preserve a strict linear null.	$\beta = 0$ solver output in `results/summary.json`.
REQ-V3-03	Verify analytic mean by numerical quadrature.	`results/quadrature_convergence.csv`.
REQ-V3-04	Verify conservative FV implementation.	Per-step mass residual in summary and documented scheme.
REQ-V3-05	Parameterize sensitivity before measurement.	`results/sensitivity.csv`; Figure 3.
REQ-V3-06	Distinguish gross Darcy flux from pore speed.	Defined in Section 4.
REQ-V3-07	Do not infer heat-flux change from $\rho c_p q \Delta T$ alone.	Section 9: exact 1D mean thermal solution.

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ID	Requirement	Verification / evidence artifact
REQ-V3-08	Record real-source and synthetic-source boundaries.	Evidence classes, Section 3, README and provenance.
REQ-V3-09	Require independent physical H-04 verification.	Laboratory gate G3, data dictionary and locked protocol.
REQ-V3-10	Terminate unsupported global thermal claims.	G6 and planetary energy comparison.

14.1 Parameter and structural uncertainty

The synthetic curves are conditional on *assumed* k_0, β, C, A, μ, L. They possess numerical precision but *no empirical confidence interval*. A future measured-flux estimate requires uncertainty propagation from calibration covariance, pressure-sensor bias, hysteresis, temperature-dependent viscosity, spatial heterogeneity, wellbore effects, fluid compressibility, leakage and model structural alternatives. For small βA, fractional sensitivity of $\overline{q}$ approximately follows $\delta k_0/k_0 + \delta\beta/\beta + 2\delta A/A - \delta\mu/\mu - \delta L/L$, but covariance and sign-changing β prohibit naive Gaussian propagation near zero.

15. Failure modes and stop rules

Failure mechanism	False interpretation at risk	Mandatory control
Pumping actuator bias	Nonzero averaged flux induced by controller, not geology.	Independent two-end flow sensing; randomized actuator sequence.
Reservoir compliance	Temporary net volume caused by tubing/connector elasticity.	Calibrate dead volume; reconcile cyclic storage after steady state.
Thermal expansion	Temperature cycle mimics a fluid volume signal.	Isothermal hydraulic phase; thermal controls and correction.
Unmeasured DC gradient	Background aquifer flow called “rectification”.	Long pre/post baselines and reversed boundary setup.
Sensor timing/phase error	Apparent nonlinear harmonic transfer.	Traceable time base, resampling audit and immutable raw data.
Wellbore storage/skin	Formation permeability inferred from distorted well head.	Explicit borehole transfer model and paired formation sensors.
Implicit energy source	“Energy generation” inferred from forced transport.	Measure pressure work, reservoir pumping and all heat boundaries.
Double counting during upscaling	Local heat redistribution mistaken for additional global energy.	Signed flux integration with boundary closure; no blanket multiplication.
Selective reporting	False positive after scanning frequencies/sites.	One primary hypothesis, blinded holdout, independent replication.

Stop H-04 for a particular material if the independently measured $k(p)$ does not support a predicted mean above calibrated sensor/leakage limits, or if the signed cycle mean remains consistent with null across preregistered replications. Stop the claim of H-05 if calorimetric change fails against conduction-only and no-flow controls. Stop H-08/global claims if integrated uncertainty includes negligible or sign-cancelling effect. Rejecting the candidate GRM law does not invalidate established tides, poroelasticity or heat transport.

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16. Energy closure and planetary limits

The model does not create heat or hydraulic work. The oscillating pressure reservoir performs work $W_A = \langle p_L q_{\text{in}} - p_R q_{\text{out}} \rangle$ per unit area at the boundaries; in the quasi-steady illustration it is about 5.025×10^{-4} W/m^2. In the transient system, storage and dissipation additionally require integration across the actual trajectory. H-05 heat advection redistributes pre-existing thermal energy; it does not directly augment planetary heat production.

For scale orientation only, the published global internal surface heat-flow estimate is 47 ± 2 TW [R10]; this corresponds to approximately 0.092 W/m^2 if spread over Earth's surface. Earth absorbs roughly 240 W/m^2 of solar power globally on average [R11]. The derived local example $\Delta J \approx 0.00310$ W/m^2 cannot be compared to either global mean as if it acts across the planet. Climate relevance would require causal, spatially signed flux integration with bounds on affected area, duration, heat source and eventual radiation to space.

> **Explicitly unsupported:** a global petroleum cooling loop; hydrocarbon extraction as a demonstrated major direct planetary warming cause via reduced subsurface coolant circulation; a single universal "pulse" transporting one identical form of energy through the Sun, ocean, rock, vegetation and human metabolism. These are excluded per PCPT v2.1 and remain excluded in v3.0.

17. Reproducibility package and integrity

The companion folder contains an executable Python source file, a fixed parameter dictionary, time-series CSV, sensitivity CSV, convergence CSV, JSON numerical summary, four figures and an input schema. The model requires NumPy, SciPy and Matplotlib. Run:

```
python -m pip install -r requirements.txt
python grm_benchmark.py
```

Record the operating system, Python and package versions, compute SHA-256 checksums, and compare the emitted values with `results/summary.json` using appropriate floating-point tolerances. Computational precision alone is not evidence of sensor accuracy, geological representativeness or independent scientific validation. The accompanying README distinguishes actual program output from literature evidence.

17.1 Suggested field dataset contract

Each observational record must contain `timestamp_utc`, `site_id`, `instrument_id`, `observable`, `value`, `si_unit`, `uncertainty_1sigma`, `quality_flag`, `calibration_id` and `raw_source_sha256`. Each operational event must carry action, start/end times, operator, baseline pressure, test set-point and independent calibration. Time-lag corrections and rejected values belong in separately versioned transformations; raw originals must not be overwritten. Real observations are never merged with `SYNTHETIC` records without an explicit origin column.

18. Research stages and publication criteria

1. **Phase A — Model validation preparation.** Complete constitutive characterization of actual porous/rock samples and quantify physically realistic imposed pressure transfer. Replace demonstration values with measurements and uncertainty bounds.
2. **Phase B — Laboratory H-04.** Build instrumented mass-conserving column; preregister experiments; investigate baseline, amplitude, frequency, reversal, tracer and leak control; conduct a separate blind replication.
3. **Phase C — Thermal H-05.** Validate transient coupled energy model with independently measured conductivity, heat capacity and thermal boundary conditions; conduct a paired no-flow control.
4. **Phase D — Field H-03/H-04.** Select sites with minimal operational confounders and independently measured tidal-loading transfer; use multi-month monitoring with withheld confirmation data.
5. **Phase E — Regional constraint.** Only if C and D pass, map spatial domain of effects and derive an uncertainty-

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bounded, signed energy and fluid mass budget. A null or negligible regional estimate is a publishable finding.

A journal submission should carry (i) code and versioned environment, (ii) calibration and independently auditable data provenance, (iii) a clearly separated synthetic benchmark, (iv) field/lab statistical analysis, (v) complete null results, and (vi) an appropriately bounded novelty statement referencing earlier tidal groundwater circulation work.

19. Final scientific conclusion

What v3.0 establishes. Under the declared boundary conditions and the hypothetical $k(p) = k_0 e^{\beta p}$ law, zero-mean boundary-pressure oscillations can yield a nonzero time-mean, mass-conserving Darcy flow. The analytic Bessel-function result and the implicit transient finite-volume solution agree to numerical precision; a linear control produces a zero mean. A separately stated mean-flow thermal approximation yields a small, local conditional change in total heat flux.
What v3.0 does not establish. It supplies no new natural permeability measurements, no experimentally demonstrated GRM in a geological specimen, no newly analysed field datasets, no proof that Earth tides reproduce the model's prescribed pressure amplitude, no full time-dependent thermal validation and no global planetary climate mechanism. Claims H-04 and H-05 therefore remain hypotheses pending controlled independent experiments.

20. References and source provenance

R1. Tsinderhoz, V. (2026). *Planetary Coupled Pulse Theory: Research Baseline v1.0*. Author manuscript, 28 July 2026, 131 pp. [User-supplied primary source].

R2. Rojstaczer, S., and Agnew, D. C. (1989). The influence of formation material properties on the response of water levels in wells to Earth tides and atmospheric loading. *JGR: Solid Earth* 94(B9), 12403–12411. doi:10.1029/JB094iB09p12403.

R3. Mango, A. J., Schmeeckle, M. W., and Furbish, D. J. (2004). Tidally induced groundwater circulation in an unconfined coastal aquifer modeled with a Hele–Shaw cell. *Geology* 32(3), 233–236. doi:10.1130/G19922.1.

R4. Tyler, R. H., Maus, S., and Lühr, H. (2003). Satellite observations of magnetic fields due to ocean tidal flow. *Science* 299, 239–241. doi:10.1126/science.1078074.

R5. Bastias Espejo et al. (2022). Groundwater Responses to Earth Tides: Evaluation of Analytical Solutions Using Numerical Simulation. *JGR: Solid Earth* 127, e2022JB024771. doi:10.1029/2022JB024771.

R6. Valois, R., Rivière, A., Vouillamoz, J.-M., and Rau, G. C. (2024). Analytical solution for well water response to Earth tides in leaky aquifers with storage and compressibility in the aquitard. *Hydrol. Earth Syst. Sci.* 28, 1041–1054. doi:10.5194/hess-28-1041-2024.

R7. Petit, G., and Luzum, B. (eds.) (2010). *IERS Conventions (2010)*. IERS Technical Note 36. https://www.iers.org/iers/en/publications/technicalnotes/tn36.

R8. NOAA CO-OPS. *Tides and Currents: Web Services / Data API*, official documentation. https://www.tidesandcurrents.noaa.gov/web_services_info.html.

R9. USGS. *Groundwater Levels Service documentation* and Water Data for the Nation. https://waterservices.usgs.gov/docs/groundwater-levels/groundwater-levels-details/.

R10. Davies, J. H., and Davies, D. R. (2010). Earth's surface heat flux. *Solid Earth* 1, 5–24. doi:10.5194/se-1-5-2010.

R11. NASA Earth Observatory (2009). Climate and Earth's Energy Budget. https://science.nasa.gov/earth/earth-observatory/climate-and-earths-energy-budget/.

R12. Biot, M. A. (1941). General theory of three-dimensional consolidation. *J. Appl. Phys.* 12(2), 155–164. doi:10.1063/1.1712886.

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R13. Tsinderhoz, V. (2026). *Planetary Coupled Pulse Theory: Corrected Scientific Edition v2.1*. Author manuscript, 28 July 2026, 144 pp. [User-supplied primary source].

Provenance rule: R1 and R13 define the source architecture, not a demonstrated empirical GRM. R2–R6 are external scientific prior-art; R7–R9 specify potential observational ingress; R10–R11 are contextual scales; R12 underlies poroelastic theory but is not solved fully in the reduced model. External reference details were independently checked against journal, USGS, IERS, NOAA and NASA entries in preparation of this edition.

A. Equation passports and dimension checks

G1: Darcy response. In $q = -(k/\mu)\nabla p$, $[k/\mu] = \mathrm{m}^2/(\mathrm{Pa\,s})$ and $[\nabla p] = \mathrm{Pa/m}$, yielding $[q] = \mathrm{m/s}$. Domain: water-like saturated single phase, horizontal 1D interval, prescribed gauge-pressure boundaries. Known failure: strong fracture hysteresis, free-surface, gas and coupled deformation.

G2: Storage balance. $[C\partial_t p] = \mathrm{Pa}^{-1} \cdot \mathrm{Pa/s} = \mathrm{s}^{-1}$ and $[\partial_x q] = \mathrm{s}^{-1}$. The conserved quantity is a linearized fluid-volume equivalent per bulk volume; no compressible fluid mass equation is solved. Known failure: large deformation, state-dependent storage and multi-constituent mixtures.

G3: Kirchhoff transform. $[F] = [kp/\mu] = \mathrm{m}^2/\mathrm{s}$; thus $[\partial_x F] = \mathrm{m/s}$. F requires positivity of k, which holds under the assumed exponential. Known failure: constitutive mismatch and nonlocal memory.

G4: Heat transfer. $[\rho c_p q\, dT/dx] = \mathrm{J/(m^3K)} \cdot \mathrm{m/s} \cdot \mathrm{K/m} = \mathrm{W/m^3}$ and $[\lambda\, d^2T/dx^2] = \mathrm{W/m^3}$. The heat boundary flux has units $\mathrm{W/m^2}$. The mean-flow screen is not a transient coupled thermal model.

B. Convergence and reproduction acceptance checks

- Run `grm_benchmark.py`; inspect `results/summary.json` for reference parameters and computed outputs.
- Confirm that $\beta = 0$ yields zero analytical and numerically negligible mean; $\beta \to -\beta$ reverses the sign of the mean.
- Verify agreement between time-grid quadrature and the Bessel-function expression at increasing samples per period; compare the CSV.
- Verify finite-volume mean outlet and inlet agreement after ten periods. Confirm per-step storage change equals boundary inflow–outflow within nonlinear solve residual.
- Inspect four figures and repeat with altered A and β; record any drift/instability rather than hiding it.
- Do *not* transfer conditional outputs into field-data tables. Parameter adjustment in the confirmatory set invalidates a preregistered result.

C. Executive conclusion in Russian / Краткое заключение

Что сделано. В редакции PCPT v3.0 центральная гипотеза геологической ректификации (GRM) переведена в конкретную проверяемую постановку: однофазное течение по Дарси, зависимость проницаемости от давления, периодические граничные условия и закон сохранения массы. Выведено аналитическое выражение среднего потока; выполнен расчёт нестационарной модели конечных объёмов и проверен линейный контроль.

Численный результат. При *условно заданных* параметрах расчёт даёт средний поток 5.0125×10^{-9} м/с; модель с постоянной проницаемостью даёт среднее, практически равное нулю. Выполнена оценка локального теплового следствия через уравнение адвекции и теплопроводности.

Научный статус. Это результаты воспроизводимого *синтетического моделирования*, а не измерения в реальной породе. Новых лабораторных и полевых данных не получено. Существующие работы о приливной циркуляции подземных вод показывают, что подобный класс физических процессов уже известен; новизна PCPT должна доказываться конкретными параметрами, количественным прогнозом и независимым экспериментом.

Следующий обязательный этап. Проверить зависимость проницаемости от давления на реальном

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образце, организовать двусторонний учёт потоков и трассерный баланс, затем независимо подтвердить перенос тепла. До этого гипотезы H-04 и H-05 не считаются эмпирически подтверждёнными. Глобальное охлаждение Земли нефтью и климатически значимый эффект нефтедобычи данным документом не доказаны.

End of PCPT v3.0 Research Baseline

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